

Physics-Informed Deep Learning Methods for Airfoil Flow Prediction

Stanford CS229 Project

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1 Introduction

Simulating flow fields around airfoils normally requires numerically solving the Reynolds-Averaged Navier–Stokes (RANS) equations or performing Large Eddy Simulations (LES). However, methods as such are computationally expensive. High-fidelity numerical solvers like OpenFOAM may take tens of seconds to compute a single steady-state solution, making them unsuitable for tasks requiring rapid evaluation such as airfoil design optimization or flow control. Recent advances in data-driven modeling have demonstrated that convolutional neural networks (CNNs) can learn mappings from boundary conditions—such as airfoil geometry and freestream velocity—to full pressure and velocity fields. Once trained, these models intake boundary conditions (airfoil shape and flow conditions) and can predict corresponding velocity field and pressure field within milliseconds, achieving orders-of-magnitude speedup compared with traditional CFD solvers while maintaining good accuracy.

This project aims to reproduce and extend the deep learning–based RANS inference framework proposed by Thuerey et al. (Thuerey et al. (2020)), exploring how neural network can accelerate aerodynamic simulation while retaining physical consistency by incorporating physics loss. We start with a purely data-driven framework as the baseline case, and then incorporate physics prior into the training process. By adding a physics-informed loss term in the loss function, we observe improvement in overall prediction accuracies and better physical consistency in the flow prediction.

2 Related Work

Deep learning has been widely applied to surrogate modeling of fluid dynamics. A representative example is the work of Thuerey et al., who use a U-Net–based convolutional neural network to directly infer RANS pressure and velocity fields from boundary-condition inputs. Their results show that supervised CNNs can approximate flow fields with decent accuracies, motivating a series of later studies on flow-field super-resolution Wang et al. (2021b), autoencoder-based reduced models Wang et al. (2021a), and generative approach for flow field predictions Liu and Thuerey (2024).

An alternative line of research focuses on physics-informed neural networks (PINNs) Raissi et al. (2019), introduced by Raissi and Karniadakis. PINNs incorporate the Navier–Stokes equations directly into the loss function, allowing models to learn from sparse or unlabeled data by penalizing PDE residuals. PINNs excel in inverse problems and scenarios with limited supervision, but they often face optimization challenges when solving stiff equations, high-Reynolds-number flows, or large-resolution domains.

Our work builds on the supervised-learning framework of Thuerey et al., while drawing inspiration from physics-informed methods. Inspired by the PINN method, we introduce a loss modification based on incompressible mass conservation to enhance prediction accuracy. We hypothesize that this additional PDE-based physics loss can serve as a regularizer for the training process and lead to more physically consistent predictions.

3 Dataset and Features

3.1 Data Generation and Dataset

Our project uses the publicly released *Deep-Flow-Prediction* dataset from Thuerey et al. (2020) for RANS flow fields around 2D airfoils. The original dataset is generated using a numerical solver called OpenFOAM, where steady-state solutions are computed over 1505 airfoils from the UIUC database with Reynolds numbers $Re \in [0.5, 5]$ million and angles of attack between -22.5° and 22.5° .

Each OpenFOAM solution is originally computed on a body-fitted triangular mesh. For use in the neural network, the flow field surrounding the airfoil is extracted and resampled onto a fixed Cartesian grid of size 128×128 . The resampling window spans a $2 \text{ unit} \times 2 \text{ unit}$ square domain centered on the airfoil, where one unit corresponds to the chord length. This region is large enough to capture the flow details around an airfoil. The dataset is stored in .npz format; each sample contains six channels:

$$d = [u_x^{\text{in}}, u_y^{\text{in}}, \phi, p^{\text{gt}}, u_x^{\text{gt}}, u_y^{\text{gt}}],$$

where ϕ denotes the airfoil mask. The first three channels encode boundary conditions, while the last three represent ground-truth pressure and velocity fields. The data set is divided into training and validation sets. Testing is performed on a separate set of previously unseen airfoil shapes. In our experiments, we use 6400 training samples, with 6000 for training and 400 for validation, and additional 90 samples for testing.

3.2 Preprocessing and Normalization

We follow the following preprocessing and normalization procedures before training:

Pressure offset removal. Since only pressure gradients influence incompressible RANS solutions, the absolute pressure level is arbitrary. Therefore, the dataset removes the per-sample pressure mean and eliminates pressure values inside the airfoil mask:

$$p \leftarrow p - \text{mean}(p), \quad p \leftarrow p(1 - \phi).$$

Dimensionless normalization. All target quantities are nondimensionalized using the magnitude of the freestream velocity. Horizontal and vertical velocities are normalized by the magnitude of the freestream velocity while pressure is normalized by the dynamic pressure:

$$\tilde{u} = \frac{u}{\|v_\infty\|}, \quad \tilde{p} = \frac{p}{\|v_\infty\|^2}.$$

This removes the quadratic scaling in pressure and reduces variance across the dataset.

Scaling to $[-1, 1]$. After nondimensionalization, each channel is normalized using fixed global maxima computed from the training dataset:

$$\tilde{p}/4.65, \quad \tilde{u}_x/2.04, \quad \tilde{u}_y/2.37,$$

This produces scaled values in $[-1, 1]$ and improves gradient stability during optimization. Although the freestream velocity is spatially constant, encoding it as full grids ensures that convolutional layers can access boundary-condition information at all spatial locations. At test time, the same normalization constants computed from the training set are reused, ensuring consistent scaling between training and evaluation stages.

4 Methods

4.1 Baseline: Purely Data-Driven Learning

Our baseline follows the fully data-driven supervised convolutional architecture introduced by Thuerey et al. (2020). The model takes as input a three-channel tensor representing the freestream velocity components and the airfoil mask $[u_x^{\text{in}}, u_y^{\text{in}}, \phi]$, and predicts the corresponding steady-state RANS fields (p, u_x, u_y) on a 128×128 grid.

The backbone of the network is a U-Net–style encoder–decoder with skip connections. The encoder progressively downsamples the spatial domain through convolutional blocks, enabling the model to capture global flow features such as separation and recirculation regions. The decoder then upsamples the latent representation using bilinear upsampling and transposed convolutions, while skip connections preserve high-frequency geometric information along the airfoil surface.

Formally, the supervised learning problem is posed as a pixelwise regression:

$$\min_{\theta} \mathcal{L}_{L1} = \frac{1}{N} \sum_{i=1}^N \|f_{\theta}(x_i) - y_i\|_1,$$

where x_i denotes the three input channels and y_i the ground-truth solution fields. The ℓ_1 loss is chosen following Thuerey et al. (2020), as it encourages sharper velocity and pressure predictions compared to ℓ_2 loss.

4.2 Physics-Informed Learning

While the purely data-driven baseline model achieves good predictive accuracy, it does not explicitly encode the governing physical laws. In our setting, the flow fields are governed by the steady Reynolds-Averaged Navier–Stokes equations closed with the Spalart–Allmaras turbulence model. The governing equations consist of the incompressible continuity condition

$$\nabla \cdot \mathbf{u} = 0,$$

and the RANS momentum equation

$$(\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p = \nabla \cdot [(\nu + \nu_t) (\nabla \mathbf{u} + (\nabla \mathbf{u})^T)],$$

where ν is the molecular viscosity and ν_t is the eddy viscosity predicted by the SA turbulence model.

Although these equations fully describe the underlying physics, directly enforcing the second momentum residual in a learning framework is challenging. The nonlinear advection term, the eddy-viscosity-dependent diffusion operator, and the strong dependence of ν_t on wall distance make the momentum residual highly sensitive to spatial discretization. Since our dataset is resampled onto a uniform 128×128 Cartesian grid—rather than the original body-fitted mesh—the finite-difference approximation of the momentum equation introduces numerical artifacts near the airfoil surface and leads to unstable or misleading gradients during training.

For these reasons, we adopt a lightweight physics-informed strategy and apply physical regularization only through the incompressible continuity condition. Specifically, we penalize the divergence of the predicted velocity field $\mathcal{L}_{\text{phys}} = \lambda_{\text{div}} \|\nabla \cdot \hat{\mathbf{u}}\|_1$,

which is numerically stable, inexpensive to compute, and directly corrects one of the fundamental constraints of incompressible flow. Our final training objective becomes

$$\mathcal{L} = \mathcal{L}_{L1} + \lambda_{\text{div}} \|\nabla \cdot \hat{\mathbf{u}}\|_1,$$

providing a balance between supervised learning and lightweight physics consistency while avoiding the numerical challenges associated with RANS momentum residuals on resampled grids.

5 Experiments / Results / Discussion

5.1 Experiments

During the training, We use the Adam optimizer with a learning rate of 6×10^{-4} , following the setting proposed in the original paper. A cosine-like decay schedule is applied through the `computeLR()` function, which gradually reduces the learning rate during the second half of training. The batch size is set to 10, following the same practice as the original work. Each experiment is trained for approximately 500k iterations, corresponding to roughly 800 epochs over the training set. We evaluate all models on the unseen test set of airfoils using the following metric:

$$E_{\text{rel}} = \frac{\sum |\hat{y} - y|}{\sum |y|},$$

evaluated for (1) pressure, (2) velocity magnitude, and (3) all channels combined.

To analyze the effect of physics priors, model capacity, and dataset size, we evaluate the following groups of experiments:

- **Baseline model.** Standard neural network trained purely with L1 loss.
- **Pressure-weighted loss.** As we observed in the baseline that the pressure error is relatively high, we introduce two variants with increased pressure channel loss weights $w_p = 2$ and $w_p = 4$ added to the L1 objective.
- **Physics-informed continuity loss.** Models trained with physics-informed regularization using $\lambda_{\text{phys}} \in \{2 \times 10^{-2}, 1 \times 10^{-1}, 5 \times 10^{-1}, 1\}$.
- **Large-dataset model.** Using a larger network trained on an expanded dataset to see its improvement.

5.2 Results

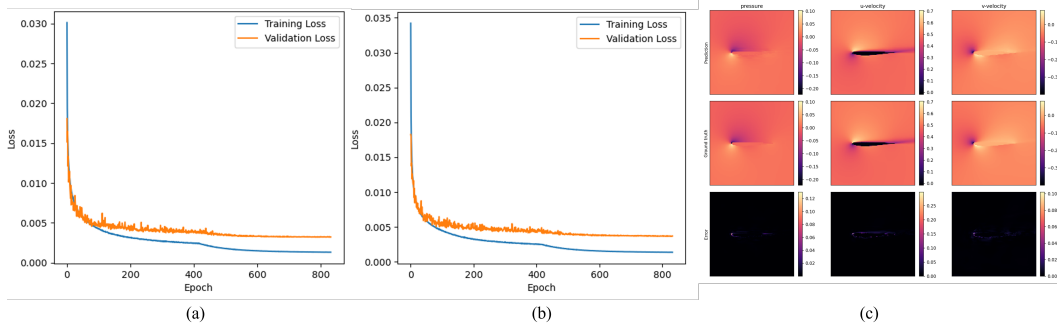


Figure 1: (a) L1 training and validation loss for baseline model (b) L1 training and validation loss for physics-informed ($\lambda = 2 \times 10^{-2}$) model (c) Example of prediction of pressure and velocity fields on an unseen airfoil.

Key results for this project are summarized in Figure 1 and Table 2. Figures 1(a) and (b) show the L1 training and validation loss curves for the baseline and physics-informed models, respectively. Overall, the two models exhibit very similar learning dynamics: in both cases, the training loss decreases rapidly during the initial epochs and then continues to decay more gradually as optimization progresses. The validation loss follows a similar trend, with an early sharp reduction followed by a slow plateau. This indicates both models successively learn from the dataset and do not overfit.

Figure 1(c) presents an example of flow-field predictions for pressure and velocity. The predicted fields closely match the ground-truth RANS solutions. These results demonstrate that the trained networks are capable of producing accurate flow reconstructions on previously unseen geometries and flow conditions.

Table 2 summarizes the relative error for all model variants across pressure, velocity, and combined channels. The baseline model exhibits the highest pressure error, which motivates the exploration of pressure-weighted losses. Increasing the pressure-channel weight ($w_p = 2, 4$) reduces the pressure error, while maintaining comparable velocity accuracy. Physics-informed models further improve the velocity predictions of the predictions, though overly large λ_{phys} can degrade accuracy due to numerical inaccuracies in the divergence computation.

5.3 Discussion

As shown in Table 2, increasing the pressure-channel weight w_p in the loss function leads to a consistent reduction in pressure error, confirming that rebalancing the loss can partially compensate for the scale disparity between pressure and velocity fields. For moderate values of the continuity penalty λ , the physics-informed model achieves lower combined error than the baseline, indicating that enforcing a weak form of mass conservation helps guide the network toward more physical predictions. However, excessively large values of λ degrade performance. This trend is unsurprising:

Table 1: Relative error (%) for pressure, velocity magnitude, and all channels combined.

Model	Pressure Err (%)	Velocity Err (%)	Combined Err (%)
Baseline	13.59	2.25	2.74
Baseline + $w_p = 2$	12.78	2.25	2.70
Baseline + $w_p = 4$	12.09	2.28	2.73
Physics loss $\lambda = 2 \times 10^{-2}$	13.01	2.15	2.61
Physics loss $\lambda = 1 \times 10^{-1}$	12.82	2.17	2.65
Physics loss $\lambda = 5 \times 10^{-1}$	13.86	2.36	2.86
Physics loss $\lambda = 1$	14.22	3.05	3.51

Table 2: Relative error (%).

Model	P	V	All
Baseline	13.59	2.25	2.74
+ $w_p = 2$	12.78	2.25	2.70
+ $w_p = 4$	12.09	2.28	2.73
$\lambda = 2 \times 10^{-2}$	13.01	2.15	2.61
$\lambda = 10^{-1}$	12.82	2.17	2.65
$\lambda = 5 \times 10^{-1}$	13.86	2.36	2.86
$\lambda = 1$	14.22	3.05	3.51

the ground-truth fields are resampled from a finer, body-fitted OpenFOAM mesh onto a 128×128 Cartesian grid and therefore do not strictly satisfy pointwise mass conservation. Strong enforcement of a physics constraint that the data itself does not obey creates a mismatch between the supervision target and the imposed regularization, ultimately hurting accuracy. If the training sample are resampled to a finer mesh (e.g 256×256), enforcing physics law could lead to more significant improvements.

To further demonstrate that the physics loss improves both physical consistency and predictive accuracy, we select a representative test case on which the baseline model fails to produce accurate predictions. Figure 2 visualizes the error contours for flow field at high angle of attack. In high-angle-of-attack flow conditions, flow separation occurs, making these regimes particularly challenging for accurate predictions. In contrast, the physics-informed model yields noticeably smaller errors. By enforcing the continuity equation, the network produces a more physically consistent flow field within the separation zone.

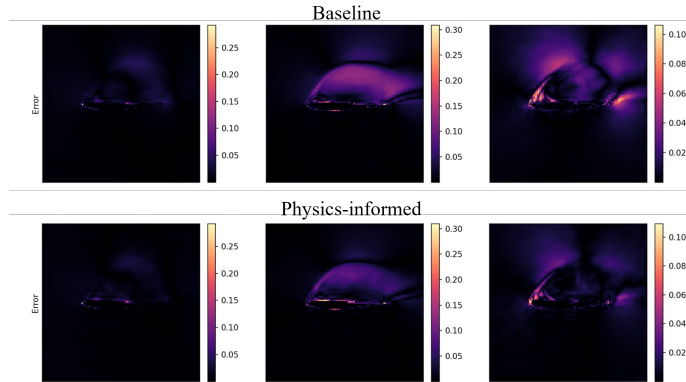


Figure 2: Error contours for a challenging high-angle-of-attack test case. The baseline model exhibits large errors in the separation region, where the flow has strong nonlinearity and recirculation. The physics-informed model significantly reduces these errors by enforcing incompressible mass conservation.

6 Conclusion / Future Work

In this project, we first reproduced the deep-learning framework of Thuerey et al. and introduced a lightweight physics-informed loss for airfoil flow prediction. The baseline achieves accurate prediction of pressure and velocity fields, while increasing the pressure-channel weight further reduces pressure error. Incorporating continuity regularization improves physical consistency and provides clearer benefits in challenging separated-flow cases, though overly strong physics weighting can degrade accuracy due to numerical inconsistencies.

For future work, we would like to implement the framework on a finer resampled data set, or employing hybrid sampling(refined sampling within the separation region) to see the full potential of physics-informed regularizer. We would also try to use different neural network structures other than u-net to explore their performances.

7 Appendices

Project code is available at:

<https://github.com/ZhiboDai804/physics-informed-deep-flow-prediction>

The u-net model structure is shown below:

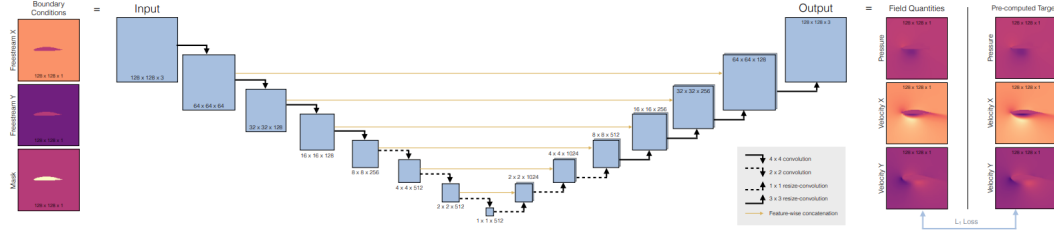


Figure 3: An overview of the neural network structure used in this projectThuerey et al. (2020)

Table 3: Network architecture of the TurbNetG generator. Here $C = 2^{\text{expo}}$ denotes the base channel count (expo = 5 in our experiments, hence $C = 32$). Each decoder block receives skip-connected features from the corresponding encoder level.

Layer	Type	Input ch.	Output ch.	Kernel	Stride	Pad	Activation / BN
Encoder							
layer1	Conv2d	3	C	4×4	2	1	None / No BN
layer2	Conv2d	C	$2C$	4×4	2	1	LeakyReLU / BN
layer2b	Conv2d	$2C$	$2C$	4×4	2	1	LeakyReLU / BN
layer3	Conv2d	$2C$	$4C$	4×4	2	1	LeakyReLU / BN
layer4	Conv2d	$4C$	$8C$	4×4	2	1	LeakyReLU / BN
layer5	Conv2d	$8C$	$8C$	2×2	2	0	LeakyReLU / BN
layer6	Conv2d	$8C$	$8C$	2×2	2	0	LeakyReLU / No BN
Decoder (Upsample + Conv2d with skip connections)							
dlayer6	Up+Conv	$8C$	$8C$	1×1	1	0	ReLU / BN
dlayer5	Up+Conv	$16C$	$8C$	1×1	1	0	ReLU / BN
dlayer4	Up+Conv	$16C$	$4C$	3×3	1	1	ReLU / BN
dlayer3	Up+Conv	$8C$	$2C$	3×3	1	1	ReLU / BN
dlayer2b	Up+Conv	$4C$	$2C$	3×3	1	1	ReLU / BN
dlayer2	Up+Conv	$4C$	C	3×3	1	1	ReLU / BN
dlayer1	ConvTranspose2d	$2C$	3	4×4	2	1	ReLU / No BN

Table 4: TurbNetG architecture ($C = 32$).

Layer	In→Out	K	S	Act/BN
Encoder				
l1	3→C	4	2	—/—
l2	C→2C	4	2	LReLU/BN
l2b	2C→2C	4	2	LReLU/BN
l3	2C→4C	4	2	LReLU/BN
l4	4C→8C	4	2	LReLU/BN
l5	8C→8C	2	2	LReLU/BN
l6	8C→8C	2	2	LReLU/—
Decoder (Upsample + Conv, skips)				
dl6	8C→8C	1	1	ReLU/BN
dl5	16C→8C	1	1	ReLU/BN
dl4	16C→4C	3	1	ReLU/BN
dl3	8C→2C	3	1	ReLU/BN
dl2b	4C→2C	3	1	ReLU/BN
dl2	4C→C	3	1	ReLU/BN
dl1	2C→3	4	2	ReLU/—

8 Contributions

Zhibo Dai: Data preprocessing, Conceptualization, Methodology, Training, Writeup.

Salavat Ishbulatov: Data preprocessing, Conceptualization, Training.

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